

2 SMART WASTE COLLECTION SYSTEM FOR A CLEANER INDIA



Waste collection and segregation is a tedious task. Collecting trash from dustbins on a regular basis requires dedicated manpower and vehicles to avoid overflow situations. A sustainable waste management solution is needed to utilise the resources and manpower efficiently. MassTrans has launched a smart waste collection system called Bigbelly in line with Clean India Mission by the government of India.

Bigbelly helps optimise garbage collection frequencies and reduces collection costs associated with it. It is a cloud-based solution that gathers, consolidates and analyses data from smart and connected stations, providing real-time alerts, live status updates and historical reports that help optimise operations and beautify public spaces.

The smart bin contains a compactor. Once garbage in the bin reaches a particular level, sensors send alerts to

the compactor to initiate the compaction (pressing down) process inside the bin. This creates more space inside the bin.

The bin has a capacity of 125 litres (un-compacted), which goes up to around 600 litres, which is around eight times higher than standard public bins. Once the smart bin is full after multiple compactations, it sends an e-mail/SMS/notification on app or cloud platform to the operating authority to collect the bin.

A single bin collects more than fifty kinds of reports. The bin is weather-proof and can even work in flooded areas with minimal damage to the bin.

The smart bin, equipped with sensors, automates the waste collection process. This saves time and resources for other productive tasks. It reduces the carbon footprint, avoids overflow problems, works on solar power, and provides connectivity through software and mobile app. It helps achieve clean, smart and cloud-connected public spaces.

The bin also has a battery inside. The solar panel charges the battery, which supplies required functionalities.

It comes in two models: one with solar panel and the other without solar panel. However, the battery is mandatory and standard for both models.

The smart waste fleet is based on compaction technology, which gives

less frequent truck rolls and optimises consumption. Multiple streams at each station encourage proper recycling and reduce greenhouse gas emissions. Each fleet has two stations with different models—high capacity (600 litres) solar-powered compacting model and standard capacity (190 litres) self-powered non-compacting model.

The system includes a set of tools for setup, management, monitoring and optimisation from either Web-based software or mobile app. The reliable and easy-to-use Web-based software provides access to real-time station status based on per cent fullness and age.

The bin provides customisable asset management and does not require any IT setup. App can be downloaded free of cost, but only the administrator of the bin can use it. The software provides actionable insights on live status and collection routines. It has not yet been developed for general public.

Sagar Pathak, product in-charge - Bigbelly, MassTrans, says, "For maintenance of the bin, owners are trained on preventive maintenance. MassTrans also helps them with maintenance."

The government is welcoming such innovations and startups to contribute towards its Clean India Mission through Make in India initiative. This will encourage businesses to come up with such innovative ideas to help make India cleaner.

—Deepshikha Shukla

3 SUSTAINABLE POWER GENERATION WITH MICRO GAS TURBINES

As environmental concerns over global warming escalate, there is a worldwide shift towards cleaner fuels for power generation for sustainable development of smart cities. Aerostrovilos Energy, an IIT-Madras incubated startup founded by

Rohit Grover, Pradeep T., Suresh M. and Satyanarayanan R. Chakravarthy in 2017, is indigenously developing micro gas turbines (MGTs) for power generation to reduce usage of such conventional fuels as coal, for lower carbon footprint.

MGTs are the smaller, simpler version of jet engines. Talking about the problem in the industry, Grover explains, "Jet engines have been used for propulsion or large power generation, but these have never been cheap enough to compete with IC engines for